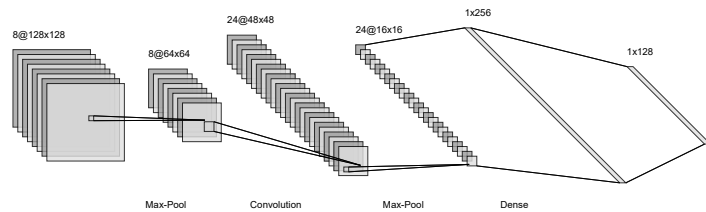




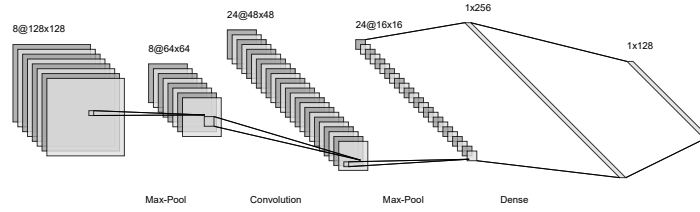
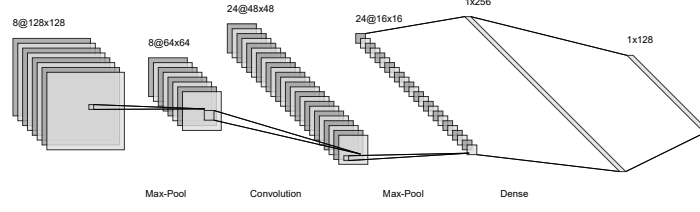
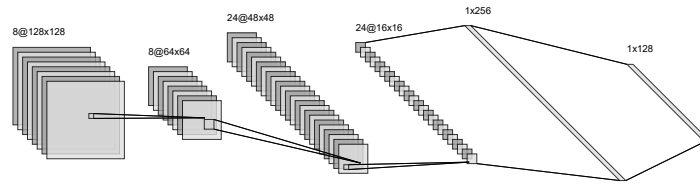
Scale to
input size



Scale to
input size



Sequential or Parallel



$n - D$ Embedding

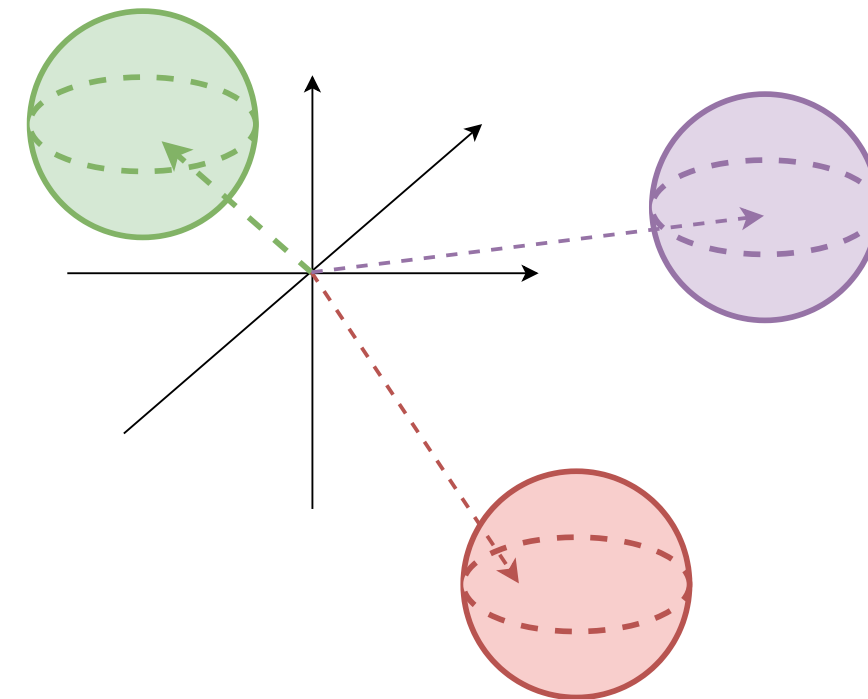
(e.g.: ArcFace 128-D)

$\begin{bmatrix} 0.1 \\ 0.35 \\ \dots \\ 0.2 \\ 0.56 \end{bmatrix}$

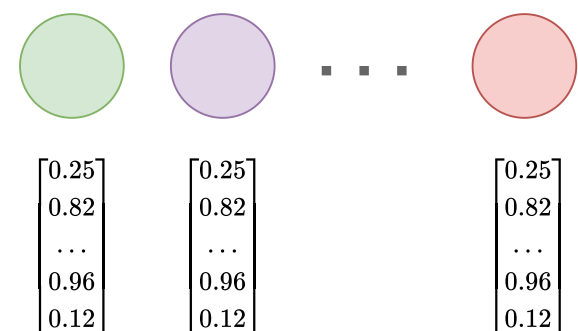
$\begin{bmatrix} 0.25 \\ 0.82 \\ \dots \\ 0.96 \\ 0.12 \end{bmatrix}$

$\begin{bmatrix} 0.37 \\ 0.49 \\ \dots \\ 0.03 \\ 0.14 \end{bmatrix}$

Train Knn for classification
of the embedding vector in a
 nD space



k clusters

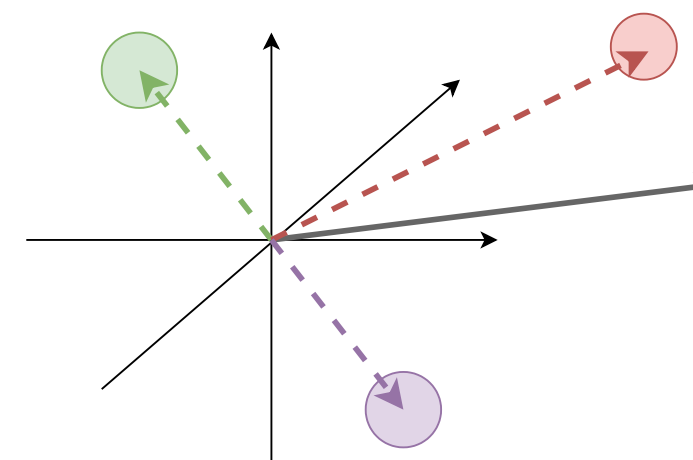


$\begin{bmatrix} 0.25 \\ 0.82 \\ \dots \\ 0.96 \\ 0.12 \end{bmatrix}$

$\begin{bmatrix} 0.25 \\ 0.82 \\ \dots \\ 0.96 \\ 0.12 \end{bmatrix}$

$\begin{bmatrix} 0.25 \\ 0.82 \\ \dots \\ 0.96 \\ 0.12 \end{bmatrix}$

Inference with
 k clusters



Metrics

$Similarity[-1; 1] = \frac{A \cdot B}{\|A\| \cdot \|B\|}$
closer 1 the better

Angle

$Euclidean[0; +\infty] = |A - B|$
the smaller the better

shortest

$Dot[0; +\infty] = A \cdot B$
the higher the better

Angle and Magnitude